



Experiment 10

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mahin-mr12 commit

9eed2b6 · 2 months ago



406 lines (406 loc) · 11 KB

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Experiment 10: Data Analysis and Visualization

1. Create numpy array to find sum of all elements in an array.

```
In [1]: import numpy as np

arr = np.array([10, 20, 30, 40, 50])

total = np.sum(arr)

print("Array:", arr)
print("Sum of elements:", total)
```

Array: [10 20 30 40 50]

Sum of elements: 150

2. Create numpy array of (3,3) dimension. Now find sum of all rows & columns individually. Also find 2nd maximum element in the array.

```
In [2]: import numpy as np

arr = np.array([[10, 20, 30],
                [40, 50, 60],
                [70, 80, 90]])

print("Array:\n", arr)

# Row-wise sum
print("Row Sum:", arr.sum(axis=1))

# Column-wise sum
print("Column Sum:", arr.sum(axis=0))

# Second maximum element
second_max = np.sort(arr.flatten())[-2]
print("Second Maximum:", second_max)
```

```
Array:  
[[10 20 30]  
 [40 50 60]  
 [70 80 90]]  
Row Sum: [ 60 150 240]  
Column Sum: [120 150 180]  
Second Maximum: 80
```

3. Perform Matrix multiplication of any 2 n*n matrices.

```
In [3]: import numpy as np  
  
A = np.array([[1, 2],  
              [3, 4]])  
  
B = np.array([[5, 6],  
              [7, 8]])  
  
result = np.dot(A, B)  
  
print("Matrix A:\n", A)  
print("Matrix B:\n", B)  
print("Matrix Multiplication:\n", result)
```

```
Matrix A:  
[[1 2]  
 [3 4]]  
Matrix B:  
[[5 6]  
 [7 8]]  
Matrix Multiplication:  
[[19 22]  
 [43 50]]
```

4. Write a Pandas program to get the powers of an array values element-wise.

Note: First array elements raised to powers from second array

Sample data: {'X':[78,85,96,80,86], 'Y':[84,94,89,83,86], 'Z':[86,97,96,72,83]}

Expected Output:

X Y Z

0 78 84 86

1 85 94 97

2 96 89 96

3 80 83 72

4 86 86 83

In [4]:

```
import pandas as pd

data = {'X': [78, 85, 96, 80, 86],
        'Y': [84, 94, 89, 83, 86],
        'Z': [86, 97, 96, 72, 83]}

df = pd.DataFrame(data)

print("Original DataFrame:\n")
print(df)

# Element-wise power (X raised to Y)
power = df['X'] ** df['Y']
print("\nX raised to power Y:\n")
print(power)
```

Original DataFrame:

	X	Y	Z
0	78	84	86
1	85	94	97
2	96	89	96
3	80	83	72
4	86	86	83

X raised to power Y:

0	0
1	4551265826121030281
2	0
3	0
4	0

dtype: int64

5. Write a Pandas program to get the first 3 rows of a given DataFrame.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 8, 20, 14.5, np.nan, 8, 10]}

```
score: [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']
```

Expected Output:

First three rows of the data frame:

attempts name qualify score

a 1 Anastasia yes 12.5

b 3 Dima no 9.0

c 2 Katherine yes 16.5

In [5]:

```
import pandas as pd
import numpy as np

exam_data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',
            'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
    'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']
}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

df = pd.DataFrame(exam_data, index=labels)

print("First three rows of the DataFrame:\n")
print(df.head(3))
```

First three rows of the DataFrame:

	name	score	attempts	qualify
a	Anastasia	12.5	1	yes
b	Dima	9.0	3	no
c	Katherine	16.5	2	yes

6. Write a Pandas program to find and replace the missing values in a given DataFrame which do not have any valuable information.

In [6]:

```
import pandas as pd
import numpy as np

df = pd.DataFrame(exam_data)

print("Before replacing:\n")
print(df)

# Replace NaN with 0
df.fillna(0, inplace=True)

print("\nAfter replacing missing values:\n")
print(df)
```

Before replacing:

	name	score	attempts	qualify
0	Anastasia	12.5	1	yes
1	Dima	9.0	3	no
2	Katherine	16.5	2	yes
3	James	NaN	3	no
4	Emily	9.0	2	no
5	Michael	20.0	3	yes
6	Matthew	14.5	1	yes
7	Laura	NaN	1	no
8	Kevin	8.0	2	no
9	Jonas	19.0	1	yes

After replacing missing values:

	name	score	attempts	qualify
0	Anastasia	12.5	1	yes
1	Dima	9.0	3	no
2	Katherine	16.5	2	yes
3	James	0.0	3	no
4	Emily	9.0	2	no
5	Michael	20.0	3	yes
6	Matthew	14.5	1	yes
7	Laura	0.0	1	no
8	Kevin	8.0	2	no
9	Jonas	19.0	1	yes

7. Create a program to demonstrate different visual forms using Matplotlib.

In [7]:

```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([1,2,3,4,5])
y = np.array([10,20,25,30,40])

# Line Plot
plt.plot(x,y)
plt.title("Line Plot")
plt.show()

# Bar Chart
plt.bar(x,y)
plt.title("Bar Chart")
plt.show()

# Scatter Plot
plt.scatter(x,y)
plt.title("Scatter Plot")
plt.show()

# Pie Chart
plt.pie(y, labels=x, autopct='%1.1f%%')
plt.title("Pie Chart")
plt.show()
```

ModuleNotFoundError

Traceback (most recent call last)

Cell In[7], line 1

```
----> 1 import matplotlib.pyplot as plt
      2 import numpy as np
      4 x = np.array([1,2,3,4,5])
```